

Research

Sustainable environmental design: evaluating the integration of sustainable knowledge in Saudi Arabian architectural programs

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Abstract

By aligning Saudi Vision 2030 with the UN's SDGs, this study highlights the significant role of integrating sustainable environmental design knowledge into architectural education in Saudi Arabia. Viewing this integration as a moral obligation, it emphasizes equipping future architects with the skills to create environmentally responsible and resource-efficient buildings, fostering a sustainable and resilient built environment. This paper aims to examine the degree of integration of sustainability knowledge and assess the structural models incorporated into the architectural programs of six Saudi Arabian universities. Adopting a quantitative approach with qualitative insights, this research focuses on universities certified by the National Architectural Accrediting Board (NAAB) or accredited by recognized national and international bodies. This study is limited to the meso-level (curricular management) and includes structural models, course descriptions, learning outcomes, course sequences, and alignment with program visions and missions. A comprehensive review was conducted on all courses related to sustainability, collecting data from program websites. The main variables analyzed included sustainable environmental knowledge, total contact hours dedicated to sustainability courses, distribution across various course types, and patterns over a 5-year period. The units of analysis encompass program plans, vision and mission statements, learning outcomes, course syllabi, and descriptions. The findings reveal significant disparities in sustainability emphasis among programs, with Imam Abdul Rahman Al-Faisal University dedicating 42% of contact hours to sustainability, compared to 25 to 11% at other universities. This study highlights the need for integrating sustainability across design studio levels to provide students with practical skills across disciplines. Most programs use a "linear/parallel" or "partially integrated" model, requiring sustainability principles in graduation projects. However, elective sustainability courses are unpopular and do not adequately address curriculum gaps. This study highlights the necessity of further investigating the effectiveness of incorporating sustainability knowledge in architectural programs. Exploring specific areas for future research to enhance the integration of sustainability in architectural education is also recommended.

Keywords Sustainability knowledge · Integration · Architectural design · Education · Structure model

1 Introduction

For thousands of years, architecture has shaped our lives and continues to do so. Before the widespread use of mechanical heating and cooling systems, architecture was often tailored to the local climate and terrain until the last century. Today, this multifaceted field encompasses design, building construction, mechanical systems, and social sciences. As Rapoport noted, architecture cannot simply be a form or something designed to "impress the populace"

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[1]. Modern architecture must address diverse solutions to meet both physical and psychological needs. The primary goal should be creating a good living environment, with resource conservation playing a key role in achieving this.

For more recent concerns, the construction sector plays a pivotal role in the economic landscape but also significantly impacts the environment. Globally, the industry is a major consumer of energy and materials and a significant source of pollution [2]. Construction involves numerous participants worldwide and encompasses various manufacturing and service sectors, including those for building materials and systems [2]. Notably, buildings consume substantial amounts of electricity, and architects play a key role in mitigating environmental damage. In the U.S., for instance, buildings use approximately 76% of the country's electricity and 40% of its total energy consumption [3]. Construction materials also contribute to global waste, accounting for 40–50% of all materials used in building products worldwide, amounting to three billion tons annually [4]. These materials are responsible for 56% of the carbon footprint of the construction and building sectors [5]. In Europe, construction and demolition waste, including materials such as wood, ceramics, and concrete, accounts for approximately 46% of the total waste [5, 6]. Recognizing these environmental challenges, building industry professionals, particularly architects, are increasingly focusing on sustainable practices early in the building design process [7].

Due to the increasing focus on resource conservation, architects are now required to design according to performance standards such as the Building Research Establishment's Environmental Assessment Method (BREEAM) and the U.S. Green Building Council's Leadership in Energy and Environmental Design (LEED). These standards address various aspects, including energy consumption, water conservation, daylighting, and indoor air quality [8]. Consequently, architects need new knowledge bases that integrate quality of place with environmentally responsive strategies, including 'green' architecture. Arguably, as buildings are designed differently, the education of architects must also evolve to meet these new demands.

As architects adapt to these evolving standards, sustainability in architecture becomes essential, balancing human needs with environmental conservation to ensure a healthier planet for future generations. This concept encompasses economic, environmental, and social dimensions, aiming to meet present needs without compromising those of future generations [7]. The pressing demand for sustainability spans numerous sectors, with a critical focus on the environment. Key environmental principles include addressing climate and comfort issues, energy use, water resources, material use, health and safety, and urban development. Integrating these principles into design practices, especially sustainable environmental design, is paramount [9]. This approach ensures that human activities harmonize with nature rather than deplete nature by incorporating eco-friendly materials, energy-efficient systems, and waste reduction strategies.

Furthermore, architects play a crucial role in shaping the built environment and promoting sustainability through design. This not only addresses global warming but also aims to enhance living conditions while reducing environmental impacts [10–12]. The role of sustainability in architecture has led to the creation of buildings that positively interact with their environments [13–19]. The relationship between architecture and nature has been explored, emphasizing comprehensive sustainability beyond mere energy efficiency [20–23]. Contemporary architectural design integrates multiple disciplines to address various aspects of human needs, including energy efficiency, indoor environmental quality, and operational costs [23].

A crucial aspect of sustainable architecture is urban planning and development processes, which play a vital role in shaping sustainable cities and communities [24]. However, urban planning promotes the efficient use of resources, reduces environmental impacts, and enhances the quality of life for residents [25]. As buildings are designed differently to meet these sustainability standards, it can be argued that the education of architects must also evolve to incorporate new knowledge bases that merge quality of place with environmentally responsive strategies.

Globally, sustainable development requires collaborative efforts beyond national policies, embodying a holistic and interconnected approach [26]. Jabareen introduced a framework for sustainable development that includes elements such as natural capital and integrative management [27]. There is a consensus on how to balance environmental, sociocultural, and economic initiatives to achieve sustainability [28, 29]. This interconnected approach highlights the necessity of addressing construction waste and the role of architects in fostering a more sustainable future.

In Saudi Arabia, efforts toward sustainability are evident in initiatives such as Saudi Vision 2030, which aligns with the United Nations' Sustainable Development Goals. The building sector in Saudi Arabia consumes 29% of the country's primary energy, which is approximately 75% of the country's total electricity production, with residential buildings alone responsible for almost half of this consumption [3, 30, 31]. Additionally, buildings contribute significantly to water consumption, accounting for 17% of Saudi Arabia's total water demand. The rapid growth of the Saudi Arabian population and urban development have significantly increased waste generation, which is projected to reach 30 million tons by 2033 [32, 33]. As a result, new knowledge bases are needed for architects who merge quality of place with

environmentally responsive strategies, including ‘green’ architecture. It could be argued that if buildings are designed differently, then the education of the architect should evolve.

The importance of integrating sustainability into architectural education was emphasized by the UNESCO/UIA Charter for Architectural Education. This approach helps prepare architects to confront 21st-century challenges and promote sustainable practices through professional capacities and societal contributions [33–36]. By evolving the education of architects to include sustainability principles, we can ensure that future professionals are equipped to design buildings that harmonize with their environment, addressing both local and global sustainability goals.

Building on this foundation, this study aimed to examine the degree of integration of sustainability knowledge and assess the structural models incorporated into architectural programs based on data from six accredited architecture programs in Saudi Arabia.

2 Architectural profession in Saudi Arabia

The architectural profession in Saudi Arabia primarily aims to provide excellent service and meet the needs of the community. Globally, architectural standards and practices are regulated by various entities, such as unions, institutes, councils, or associations, depending on the region. For example, in the UK and the US, architectural graduates are required to register with jurisdictional boards, obtain licenses, and pass examinations before practising. Notable organizations such as the Architects Registration Board (ARB) in the UK and the National Architectural Accreditation Board (NAAB) in the U.S. set educational standards to ensure ongoing professional excellence, although their accreditation criteria vary [10, 35].

In Saudi Arabia, architectural firms are required to obtain licenses to operate professionally. Along with their level of expertise, their licensing status determines their ranking within the Saudi classification system for engineering and architectural practices. Generally, Saudi accreditation authorities do not have the power to directly recognize the qualifications of foreign-trained architects who wish to practice in the country. Consequently, it is crucial to administer an examination to evaluate the credentials of these international architects, maintaining high standards in the architectural field. The Saudi evaluation system, known as “Qiyas,” plans to implement a unified examination for individuals seeking professional architectural licenses. This examination process aims to train, prepare, and verify architects, ensuring that their education and qualifications are legitimate and certified. The exam is a requirement for all architects, whether they have graduated from local schools or held degrees from foreign institutions, who intend to practice architecture in Saudi Arabia [36].

2.1 Comparative analysis of international and local practices

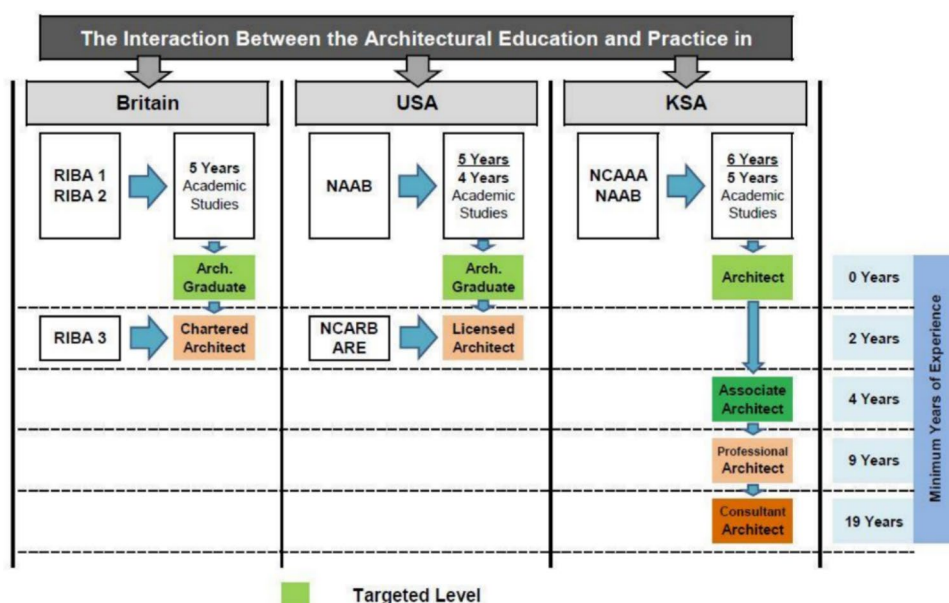
Architectural practices vary significantly across different countries, reflecting unique educational and professional standards. This section compares practices in the United Kingdom, the United States of America, and the Kingdom of Saudi Arabia, offering insights into their respective frameworks and qualifications.

In the UK, architectural education is integral to the Royal Institute of British Architects (RIBA), rooted in its 1834 Royal Charter. The RIBA examination system ensures professional standards, requiring graduates to hold RIBA-validated qualifications, gain practical experience, and obtain the RIBA Part 3 qualification. Successful completion allows registration with the Architects Registration Board (ARB) and Chartered Membership of the RIBA (Fig. 1) [37]. The RIBA also supports sustainability through its eight outcomes in the RIBA 2030 Climate Challenge, which include net zero operational and embodied carbon, a sustainable water cycle, connectivity and transport, land use and biodiversity, health and wellbeing, communities and social value, and life cycle costs. These metrics guide architects to design buildings that contribute positively to the environment and society [38].

In the U.S., two key organizations oversee architectural education and licensing: the National Architectural Accreditation Board (NAAB) accredits academic programs, while the National Council of Architectural Registration Boards (NCARB) manages licensing. The licensing process involves earning a professional degree, completing the Architectural Experience Program (AXP), and passing the Architect Registration Examination (ARE).

In Saudi Arabia, graduates can assume the title ‘architect’ upon completing a 5-year bachelor’s degree. The professional classification by the Saudi Council of Engineers recognizes architects based on experience and qualifications, ranging from Architect to Consultant Architect (see Fig. 1). Although academic accreditation is not mandatory for registration, efforts toward accreditation have begun, with examples such as King Saud University’s Department of Architecture and Building Science earning NAAB substantial equivalency accreditation [37].

Fig. 1 The interplay between architectural education and professional practice in the United Kingdom, the United States, and the Kingdom of Saudi Arabia, source from ref. [37]



Registration with the Saudi Council of Engineers is essential for practicing architects, with advancements based on experience and qualifications. However, to align with international standards and ensure the quality of architectural education and practice, it is advisable for Saudi Arabia to adopt the RIBA validation procedure used by many architecture schools worldwide. This step would enhance the credibility and professionalism of architects in the country, facilitating their integration into the global architectural community. The interaction between architectural education and practice is depicted in Fig. 1 [37].

2.2 Architectural education in Saudi Arabia: a brief presentation

During the economic boom of the 1970s and 1980s, fuelled by oil exports, Saudi Arabia witnessed the establishment of several architectural programmes. Between 1967 and 1983, five initiatives were launched [39], with King Saud University (KSU) being the pioneer in establishing an architectural department in 1967. Subsequently, the University of Dammam (UOD) in 1975, King Abdul-Aziz University (KAU) in 1976, King Fahd University of Petroleum and Minerals (KFUPM) in 1980, and Umm Al Qura University (UAU) in 1983 followed suit [39, 40], along with eight other programs later. This study focused on six programs accredited by regional or international bodies (Table 1).

Architectural programmes in Saudi Arabia, much like those elsewhere, prioritize design as the central focus. The curriculum emphasizes design education, complemented by history and theory courses. These Saudi Arabian architectural schools adopt an American paradigm inspired by Bauhaus principles, emphasizing design education, history, and philosophy. However, concerns have been raised regarding graduates' neglect of the long-term implications of their designs, reflecting the country's rapid economic and technological advancements. For instance, a number of Saudi architects are incorporating Western architectural principles, such as the use of large steel and glass structures [41].

In Saudi Arabia, higher education is under the complete authority of the Ministry of Education. Unlike the American Institute of Architects (AIA) in the United States and the Royal Institute of British Architects (RIBA) in the United Kingdom, professional organizations in Saudi Arabia do not establish standards and regulations for overseeing architectural education and practice. The absence of standardized oversight for the 13 architecture schools prompts suggestions for the establishment of professional organizations akin to the AIA and RIBA in Saudi Arabia. The implementation of an accreditation system for professional degree education is deemed essential for enhancing the value, relevance, and effectiveness of the architectural profession. Proposals include leveraging the expertise of other nations and forming an independent accrediting board in Saudi Arabia tailored to the country's specific conditions [42].

Currently, accreditation boards are not widely utilized in Saudi Arabian architectural schools. While the RIBA and NAAB validate and certify schools, they do not accredit their programs. Of the thirteen architecture programs in the country, four have received certification from the NAAB. Additionally, three programs are accredited by the National Center for

Table 1 List of Architectural programs taught in KSA, arranged by year of establishment

University	Location	Year of inauguration	Bachelor Type	International accreditation	National accreditation
KSU	Riyadh	1967	BSc. Arch	NAAB ^a	–
IAU	Dammam	1975	B. Arch	NAAB	–
KAU	Jeddah	1976	B. Arch	–	–
KFUPM	Dhahran	1981	B. Arch	–	NCAAA ^b
UQU	Mecca	1982	B. Sc. Arch	ASII ^c	–
YU	Riyadh	2001	B. Arch	–	–
BU	Baha	2005	B. Arch	–	–
EU	Jeddah	2005	B. Arch	NAAB	NCAAA
QU	Qasim	2008	B. Arch	–	–
DAU	Riyadh	2008	B. Arch	NAAB	–
JU	Jizan	2010	B. Arch	–	–
PSU	Riyadh	2011	B.Sc. Arch	–	–
KKU	Abha	2018	B. Arch	–	–

^aNational Architectural Accrediting Board^bNational Center for Academic Accreditation and Evaluation^cAccreditation Agency for Study Programmes in Engineering, Informatics, Natural Sciences and Mathematics (ASIIN)

Academic Accreditation and Evaluation (NCAAA), which applies standards for both institutions and individual programs (Table 1). However, the NCAAA does not specifically focus on architecture schools [43].

To address this gap, leveraging international expertise is essential for enhancing the quality of architectural education in Saudi Arabia, ensuring that it aligns with the country's unique circumstances. Incorporating global best practices would not only raise educational standards but also integrate crucial elements of sustainability into the curriculum. Emphasizing sustainability is vital for fostering architects who can design environmentally responsible buildings, addressing the long-term environmental challenges posed by rapid development and aligning with international sustainability goals. This step would promote higher standards and better integration into the global architectural community.

2.3 Models of the integration of environmental knowledge into architectural education

Sustainability encompasses a broad spectrum of topics relevant to the constructed environment. Therefore, it is advisable to integrate sustainable architecture education into the curriculum to offer students ample opportunities to acquire and apply this knowledge. Students currently receive instruction in sustainable architecture through a combination of theoretical and analytical approaches facilitated by appropriate pedagogical tools, techniques, simulations, measurements, and software. Exposure to sustainability issues occurs within the physical environment, both in studio lecture series and seminars. Practical application is emphasized through design studioworks, workshops, case studies, and site visits. These disciplinary domains play a crucial role in supporting the overall pedagogical process. To ensure the lasting impact of knowledge throughout the curriculum, five distinct models of program structure, as depicted in Fig. 2 and proposed by Altomonte in 2012 [44], have been suggested:

- *Parallel/linear*: Disciplinary domains run independently, preventing the integration of theoretical (nonsemplicative courses) and practical (studio/design courses) domains.
- *Partially integrated*: multiple areas are linked in delivery or, more typically, in assessment, where some modules in sustainability are taught and linked under other units while others are taught and linked as stand-alone units.
- *Fully integrated*: multiple disciplines converge around the primary core of the design studio project [44]; this structure is commonly used in design studios as a principal domain that encompasses theoretical knowledge connected to environmental effects and sustainable architecture [35].
- *Iterative model*: is of the linked phase's approach where the knowledge in sustainability is delivered in a series of domains. The components of environmental science and design studios have obvious dependencies on one another.

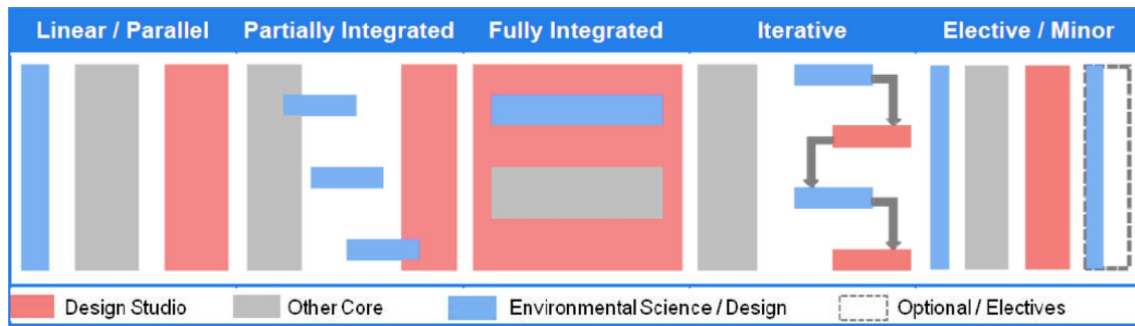


Fig. 2 The five models of the program structure [44]

- *Elective or minor models:* There are numerous electives available in the program to encourage discipline; its strength is that it offers a large variety of free choice modules that can also be structured as minor degrees to be taken from other degrees or departments in the future [12, 45].

3 Methodology

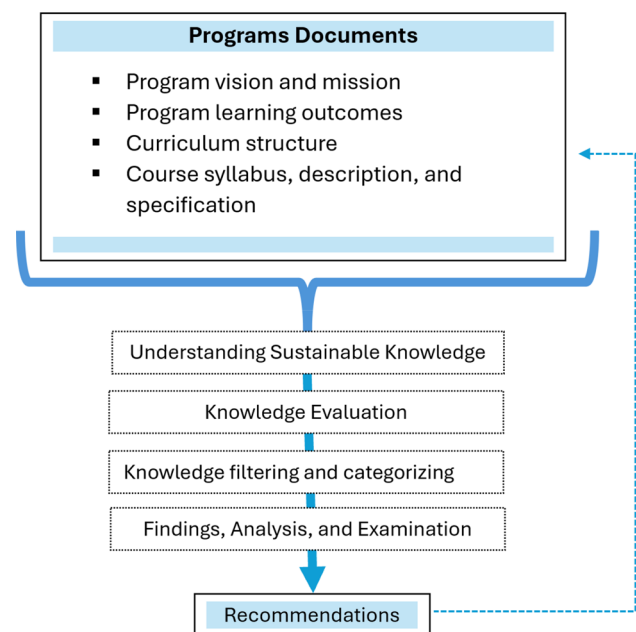
3.1 Introduction

This research aimed to examine the degree of integration of sustainability knowledge and assess the structural models incorporated into the architectural programmes of six Saudi universities. Employing a mixed-method approach, the study combines both quantitative and qualitative methodologies, with a predominant focus on quantitative analysis and minor qualitative aspects to provide context and depth. The main variables analyzed are sustainable environmental knowledge, total contact hours dedicated to sustainability courses, distribution across various course types, and patterns over a 5-year period. The units of analysis encompass program study plans, vision and mission statements, learning outcomes, course syllabi, and descriptions.

3.2 Research design

This study employs a mixed-method approach, incorporating both quantitative and qualitative methodologies, with a predominant focus on quantitative analysis to measure and evaluate the integration of sustainability knowledge in architectural programs supplemented by qualitative methods for contextual depth. The quantitative analysis aims to do the following:

1. *Measure Sustainability Knowledge:* Assess the extent and depth of sustainability knowledge integrated into the curriculum.
2. *Analyze Contact Hours:* Evaluate the total number of contact hours dedicated to sustainability-related courses, including a breakdown of hours per course and the overall emphasis on sustainability within the curriculum.
3. *Examine Course Distribution:* Investigate how sustainability content is distributed across different types of courses (e.g., core courses, electives).
4. *Programme structure models:* To examine how sustainability education has changed over the course of 5 years to identify trends and shifts in the curriculum's emphasis on sustainability. For example, as shown in Fig. 3 and suggested by Altomonte [44], we ascertain which model of integration for sustainability knowledge has been employed in each study year and over the course of the program. While the study is predominantly quantitative, qualitative aspects are incorporated to provide depth and context. This includes:
 - *Course Descriptions:* Reviewing and analyzing course descriptions to understand the content and objectives of sustainability-related courses.
 - *Vision and Mission Statements:* Examining the universities' vision and mission statements to assess their alignment with sustainability principles.

Fig. 3 Research method map

- *Learning Outcomes:* Review the stated learning outcomes for sustainability courses to determine the extent of emphasis on integrating sustainability knowledge.

This mixed-method approach ensures a comprehensive assessment of how sustainability knowledge is integrated into architectural programs, providing both broad quantitative measures and detailed qualitative insights.

3.3 Scope and limitations

The study will focus exclusively on programs certified by NAAB international certification or accredited by recognized national or international higher education-related accreditation organizations. It is also limited by the meso-level (curricular management) nature of architectural programs, examining curriculum design rather than the teaching methods or day-to-day instructional practices used in individual classrooms. The analysis encompasses structural models of the curriculum, course descriptions, learning outcomes, course sequences, and alignment with the programs' visions and missions.

3.4 Data collection

A comprehensive review was conducted on all courses related to sustainability, collecting data from program websites. This review scrutinized integration ideas, program plans, vision and mission statements, learning outcomes, course syllabi, and descriptions (see appendix A). The documents and data collection occurred in mid-2023, and the data were considered reliable and up-to-date. For example, program information, course syllabuses, descriptions, and learning outcomes are updated and often contain information about environmental conditions and human behavior issues and discuss issues of sustainability, such as energy-saving issues, the efficient use of nonrenewable resources, climate and design, water management and recycling, and material recycling and reuse. This study encountered minimal limitations concerning the availability of online data. The selected programs offered curricula on their websites, detailing course credit hours, contact hours, course descriptions, and course learning outcomes. However, most courses lacked access to detailed syllabi, which provide more comprehensive information. This approach aligns with a similar methodology used by Ismail et al. (2017) in their article, "Integrating sustainability knowledge into architectural education: UK and USA practices." [11].

3.5 Data analysis

After the collected knowledge was reviewed, evaluated and understood, the process of assessing the integration of sustainable knowledge into architectural design programs involved compiling and organizing the gathered data and information through evaluation, filtering, and categorization (Fig. 3). Sun and Hao [45] noted that knowledge needs to be evaluated to ensure that it is accurate and valuable before it can be shared. Knowledge was evaluated based on the relevance of the research goals and objectives [46].

3.5.1 Quantitative analysis

The researcher proceeded in two phases:

First phase: Measure the degree to which sustainability knowledge is available in the courses. revised this statement: Sustainable knowledge encompasses terms such as “sustainability”, “green”, “ecological”, “eco-friendly”, and “environmentally friendly”, among others, related to sustainable environmental design. These terms primarily relate to the environmental pillar of sustainability, emphasizing practices and products that minimize environmental impact and protect natural resources. The study looked at:

- The percentage of the total number of contact hours dedicated to sustainability-related courses, with a breakdown of hours per course.
- The distribution of sustainability content across core courses, electives, etc.
- Theoretical and practical courses.
- The models of program structure were built for each year of study.

Second phase: The programme courses were categorized into three types: sustainable, nonsustainable, and partially sustainable. A partially integrated sustainable course includes significant sustainability components or modules but does not focus exclusively on sustainable practices. While sustainability is integrated into the course, it is not the primary subject. The learning outcomes mention sustainability objectives alongside other core objectives. For example, an urban planning course may aim to teach students about sustainable urban development practices.

3.5.2 Qualitative analysis

The researcher conducted the content analysis, which included the following steps:

- The course descriptions were reviewed to understand the content and objectives of the sustainability-related courses.
- Assessing the alignment of vision and mission statements with sustainability principles.
- Evaluating the intended competencies and knowledge in learning outcomes related to sustainability.

During filtering and categorization, knowledge is divided into four separate groups of categories that match the usual topics covered in architecture classes and the rules set by the National Council of Architectural Registration Boards (NCARB) in the U.S. These categories include technologies and systems, skills and design, humanities and theories, and elective courses. Usually, technological and systemic aspects address knowledge about sustainability. They investigate environmental, sustainable, and ecological design ideas, such as green technologies, sustainable constructions, and building materials. Considering skills and design as the foundational pillars of architectural education, it is assumed that sustainability-related knowledge is integrated into the design studio to propel a shift toward more sustainable practices. Cultural and historical dimensions, typically covered by the humanities and theoretical disciplines, benefit from acknowledging key variations in cultural and educational contexts when exploring such topics. Additionally, elective courses contribute significantly to architectural education by offering a diverse array of topics, thereby enriching knowledge in specific areas. Furthermore, providing hands-on experience in a professional setting is deemed essential to students' architectural education.

The filtered and categorized data were discussed, analyzed, and scrutinized to formulate recommendations for improving the integration of sustainability knowledge into architectural programmes (Fig. 3).

3.6 Description of the accredited architecture programs

The six institutions detailed in Table 2 exhibit similar program durations, study plan structures, and degree prerequisites. Each of these institutions is certified by either the NAAB or the NCAAA.

3.6.1 King Saud University (KSU)

The Department of Architecture and Building Sciences laid the groundwork for King Saud University's College of Architecture and Planning in 1984 CE. This department, initially part of the College of Engineering since 1967 CE, eventually became the cornerstone of the first academic institution on the Arabian Peninsula to offer architectural education. The college encompasses two main sections: architecture and building sciences and urban planning. These disciplines are regarded as areas of human knowledge that blend art and science, exhibiting close connections to other arts and humanities. Notably, the school received substantial equivalence (SE) accreditation from the National Architectural Accrediting Board (NAAB) in the U.S. in January 2013, which was valid for 6 years [47].

3.6.2 Imam Abdulrahman Bin Faisal University (IAU)

In 1975, CE, Imam Abdulrahman bin Faisal University, formerly known as King Faisal University, inaugurated its College of Architecture and Planning. Since then, colleges have solidified their position as the nation's leading institution for studying the built environment. The Department of Architecture at the university follows an educational philosophy grounded in several principles, including the belief that an environment honoring both architectural and design ideals cultivates a conducive atmosphere for original thinking [48]. The department is dedicated to assisting students in refining their design skills, emphasizing the technical aspects of the field, and improving their communication abilities, focusing on clarity, eloquence, and professionalism.

The comprehensive 5-year degree curriculum covers not only the local environment, economic factors, and social features of the region but also delves into design, construction, theories of architecture, and engineering science. The school has undergone regular inspections by NAAB's visiting teams as of October 2022, providing international certification but not accrediting its programs.

3.6.3 King Fahd University of Petroleum and Minerals (KFUPM)

The Architecture Department, established in 1981, obtained accreditation from the NCAAA in 2015. Currently, the department provides a Bachelor's degree in Architecture (B. arch) that places a strong emphasis on architectural design, theory and history, sustainability, and urbanism. Students are also introduced to diverse areas related to design, including computer graphics, computer-aided design, structural design, human factors, and regional architecture [49].

Table 2 An overview of the KSA's accredited architecture programs, arranged by year of establishment

University	Location	Year of inauguration	Program Name	Program Length	International Accreditation	National Accreditation	Credit Hours	Status
KSU	Riyadh	1967	BSc. Arch	5	NAAB	-	170	Public
IAU	Dammam	1975	B. Arch	5	NAAB	-	170	Public
KFUPM	Dhahran	1981	B. Arch	5	-	NCAAA	185	Public
UQU	Mecca	1982	B. Sc. Arch	5	ASIIN	-	165	Public
EU	Jeddah	2005	BSc Arch	5	NAAB	NCAAA	171	Private
DAU	Riyadh	2008	B. Arch	5	NAAB	-	172	Private

3.6.4 Umm Al-Qura University (UQU)

In 1982, Umm Al-Qura University launched the Department of Islamic Architecture as part of the newly formed College of Engineering and Islamic Architecture. The program curriculum covers all aspects of architecture, with an emphasis on Islamic design principles. The department's mission is to educate students in the art and science of creating a constructed environment conducive to human flourishing and well-being. All components of the building's design are being followed in accordance with Islamic scientific principles. ASIIN granted the department's curriculum accreditation in 2018 [50].

3.6.5 Effat University (EU)

Effat University, founded in 1999 and inaugurated as a university in 2009, is dedicated to women's education in Saudi Arabia. Its 5-year bachelor of science in architecture program covers history, theory, drawing, and structures, with additional courses in technology, art, humanities, engineering, physical and social sciences, and management. The curriculum prepares graduates for professional practice locally and globally, meeting the standards of the Saudi Ministry of Higher Education, the National Commission for Academic Accreditation and Assessment (NCAAA), and the National Architectural Accrediting Board (NAAB) in the United States [51].

3.6.6 Dar Al-Uloom University (DAU)

The Architectural Engineering Department, established in 2009 within the College of Architectural Engineering and Digital Design (CADD) at Dar Al Uloom University, offers a 5-year bachelor's degree program and emphasizes creative disciplines in architecture and design, incorporating cutting-edge technology while upholding cultural heritage values and environmental considerations. Importantly, the programme received substantial equivalence with the NAAB for a period of 6 years, starting in January 2018 [52].

4 Results and discussion

4.1 General

When sustainability knowledge was examined across the four categories, it was found that the six schools that were chosen had different course structures for architectural education. Overall, the review indicated that, to date, most architecture departments in these universities provide limited sustainability knowledge in their programs, mainly focusing on theoretical principles of environmental design and sustainability. This knowledge is not required in design studio projects in the first year (the preparatory year) or second year of plan studies. Knowledge of sustainability is implemented in some schools in the design studio and in courses in the third, fourth, and fifth years, both theoretically and practically. The curriculum emphasizes design studio projects where sustainability knowledge must be translated and implemented practically in design. However, this requirement was not met in design studio projects during the first and second years. The integration of sustainability principles is included in the curriculum of certain educational institutions through the inclusion of design studios and courses in the third, fourth, and fifth years. This integration encompasses both academic and practical aspects of sustainability. Where the curriculum emphasizes design studio projects, knowledge of sustainability must be translated and applied.

The educational institutions demonstrated a range of program structures and strategies that have been used to include sustainability education in their curricula. For example, in the second and fourth years, most chosen schools of architecture used a linear program structure, where courses with sustainability-related teaching were taught independently, preventing the integration of theoretical and practical aspects. In Year 3, the integrated model of most school programs was partially integrated. This model involved the delivery of sustainability-related lectures during design studio projects, while other courses were taught in a theoretical manner. Furthermore, during the concluding year of the program, three educational institutions fully integrated sustainability knowledge into their studio projects. While two schools have chosen to provide this knowledge through theoretical courses, along with

partial integration into studio design courses, in these courses, students gain practical experience in implementing the principles and minimum requirements of sustainable design.

In Saudi Arabia's architectural curricula, design takes precedence, mirroring trends in global institutions. Historically, science, technology and system classes have been considered less crucial for enhancing students' architectural design skills. Figure 4 illustrates that the knowledge content in the selected architecture schools predominantly focuses on design and skills (53%). This emphasizes practical application in design studios and other related courses. The distribution follows humanities and theories (23%), technologies and systems (21%), and electives (3%). The study plans across these institutions allocate more contact hours to design and skills, particularly in the design studio, and less to humanities and technology courses. The design studio serves as a crucial platform for applying the knowledge gained throughout the program. Therefore, architecture schools should teach about sustainability concepts and how to use them in their design studios after covering them in courses such as building technology, systems, humanities, and theory. Even though the six schools have similar program lengths, course structures, and degree requirements (Table 2), Table 3 shows that the number of contact hours spent on each of the four types of knowledge is slightly different. Figures 5 and 6 compare the total contact hours among the six architectural programs for the four knowledge categories. Consequently, the proportion and emphasis of architectural education exhibit variability across institutions. Compared with other institutions, KSU allocates fewer contact hours to humanities and design courses, emphasizing technologies and systems (33.52%) [47]. On the other hand, UQU, the EU, and KFUPM prioritize skills and design, accounting for 58%, 57%, and 56%, respectively, more than other knowledge categories [49, 50, 53].

4.2 Integration models implemented in Saudi Arabian architecture colleges.

Figure 7 illustrates the findings from an analysis of the extent to which sustainability-related knowledge is incorporated into specific architecture programs at Saudi Arabian universities. The findings revealed variations in the proportion of sustainability-related architectural courses among the selected institutions, ranging from 11% at the University of Dar Al-Uloom to 42% at Imam Abdulrahman bin Faisal University. Notably, Imam Abdul Rahman Al-Faisal University exhibited a 42% implementation rate of sustainability knowledge in total teaching contact hours for the program, surpassing the implementation rates of other universities ranging from 25 to 11%.

In brief, no clear rules were found for how to include sustainable architecture knowledge in the programs of the schools that were examined, even though these schools are recognized by international accreditation standards such as the NAAB or local accreditation standards such as the NCAAA. The predominant focus across these schools was on skills and design (practical courses) related to sustainability within the study plans. Furthermore, there was limited interest in offering knowledge related to sustainability as elective courses, except for IAUs [48].

Fig. 4 Percentages of average contact hours for the four categories of knowledge in the six schools

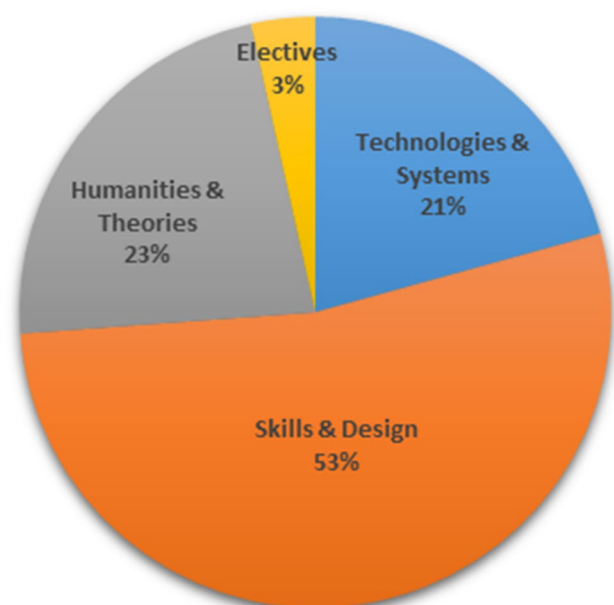


Table 3 Features of the six programs' core courses in terms of sustainability—Contact Hours (CHs)

Institution's name	Total Program Core Courses (CHs)	Technologies and systems			Skills and design			Humanities and theories			Electives courses						
		Sustain-able	Partially Sustain-able	None	Total CHs (%)	Sustain-able	Partially Sustain-able	None	Total CHs (%)	Sustain-able	Partially Sustain-able	None	Total CHs (%)				
IAU	199	0	4	38	21.11	56	8	28	46.23	3	0	50	26.63	9	9	48	6.03
KSU	179	3	0	57	32.09	28	0	53	43.32	4	4	30	20.32	10	2	32	4.28
KEUPM	160	0	8	13	16.46	10	0	80	65.82	3	3	43	13.92	3	–	33	2.53
UQU	199	2	0	30	15.76	20	0	95	56.65	6	2	44	25.62	2	–	4	1.97
UEU	196	4	0	26	15.31	10	0	101	56.63	7	6	28	21.94	6	–	45	6.12
DAU	202	0	0	47	23.27	0	20	80	49.50	2	0	53	27.23	–	–	–	–

Fig. 5 Comparing the total contact hours of core courses across four knowledge categories in six architectural programs

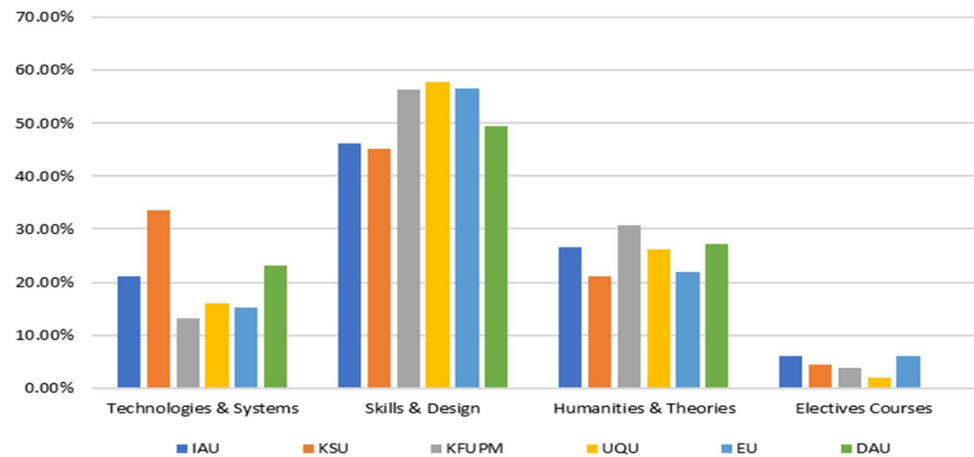


Fig. 6 Analysis of Sustainability and Partial Sustainability Ratios (Contact Hours) Across Four Knowledge Categories

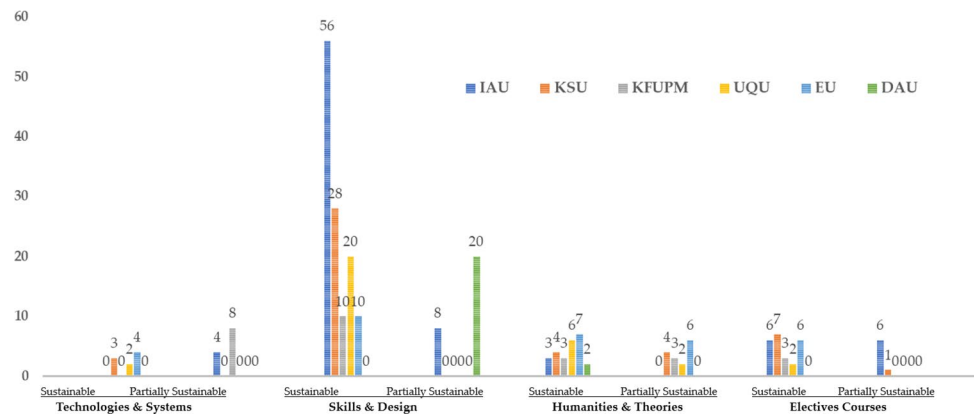


Fig. 7 The proportion of course contact hours devoted to sustainability in architectural education syllabi in Saudi Arabia

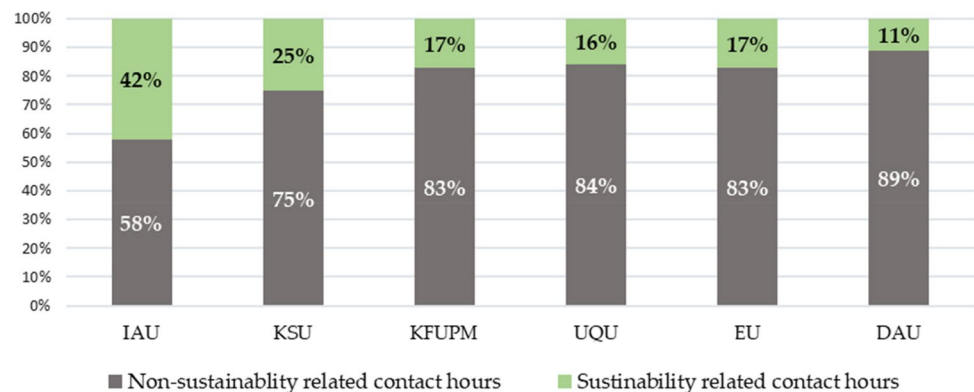


Figure 8 underscores the significance of practical courses, particularly those emphasizing skills and design related to sustainability, over theoretical courses. Notably, sustainable knowledge is taught mostly in design studios at all schools except for Effat University, where the focus is on humanities and theories in regard to implementing sustainability. Imam Abdulrahman bin Faisal University has taken the lead in terms of the number of contact hours, with 64, constituting 77% of the total sustainable knowledge allocated to design studios. Close behind are KSU, with 28 CHs (59%); UQU, with 20 CHs (62%); DAU, with 20 CHs (91%); and 10 CHs (37%) [47, 50, 52]. It is essential to note that elective courses related to sustainability are not widely integrated into three of the selected institutions, except for King Saud University, Imam Abdulrahman bin Faisal University, and Effat University. In these schools, the proportions of sustainability-focused contact hours in the overall curriculum were approximately 17%, 14%, and 9%, respectively (Fig. 9) [54–56].

Figure 9 reveals that, currently, all the selected schools of architecture except Umm Al-Qura University do not offer sustainability knowledge in Year 1, which provides a theoretical course titled “Building Materials.” All the institutions,

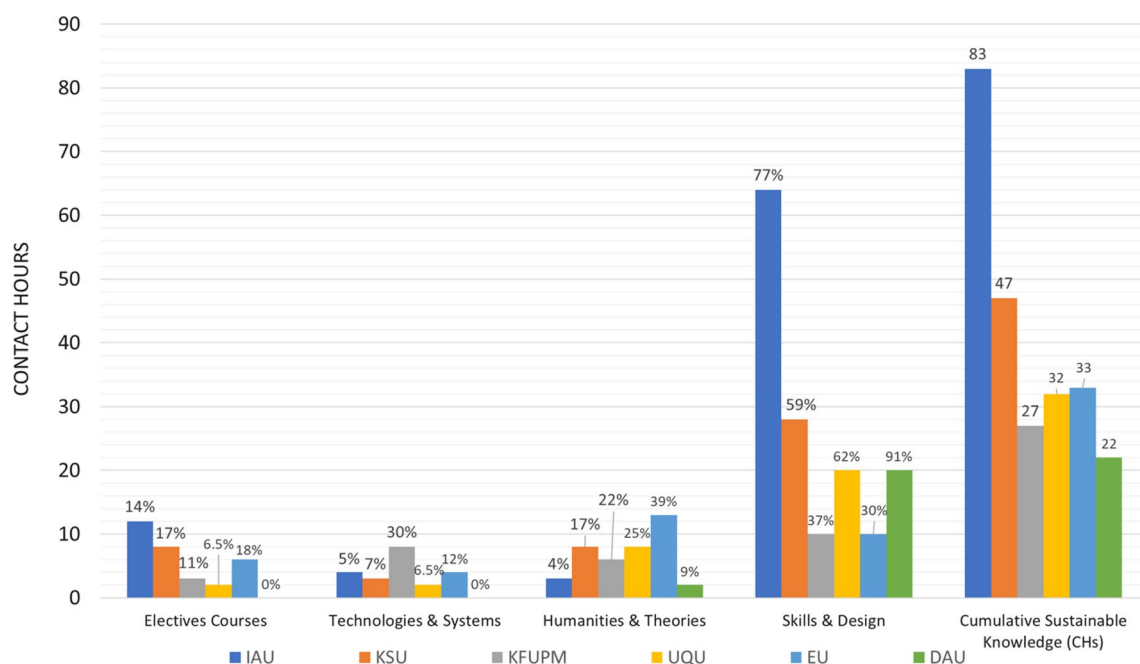


Fig. 8 The contact hours and distribution of available sustainable knowledge in the chosen schools were determined utilizing the four knowledge categories

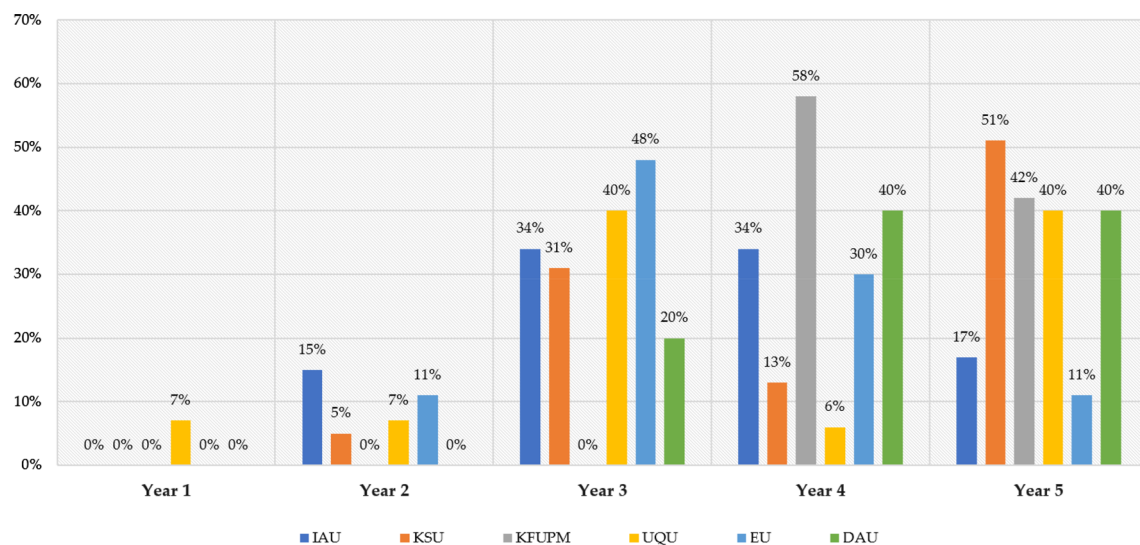


Fig. 9 Proportion of Courses (CHs) Integrating Sustainability Knowledge in the Curriculum of Chosen Architecture Schools in Saudi Arabia (KSA)

apart from KFUPM and DAU, incorporated sustainability knowledge in approximately one course during the second year, addressing both theoretical and practical aspects. Moreover, excluding KFUPM, each institution introduced between one and two sustainability-related courses in the third year. In the last year of the school's program's curriculum, almost all the institutions have placed varying amounts of emphasis on knowledge of sustainability in design studios, which requires knowledge of sustainability to be translated and applied practically. Certain academic programs offer sustainability-related electives, whereas others require sustainability-related knowledge in their core courses. There are different types of programmed structures depending on the school system and how the knowledge is taught theoretically, practically, and optionally [47–50, 52, 57].

In summary, there is currently no obvious framework for the theoretical and practical delivery of sustainable knowledge and its application in the design process. Each architecture program adopts different approaches and structures based on the college or university system, resulting in variations in the incorporation and exploration of sustainability knowledge. Notably, sustainability-related courses have been prominently featured in the last 3 years of some programs, as illustrated in Fig. 9.

According to the College of Architecture and Planning at Imam Abdulrahman bin Faisal University, 68% of all courses related to sustainability were taught in years 3 and 4 (34% each). The remaining 15% and 17% of sustainable knowledge is used in years 2 and 5 of the program study plan, respectively. Courses related to sustainability are not offered in the first year. This is achieved through the incorporation of environmental system courses and design studios [48]. No academic year is devoid of implementing one or two practical courses teaching some knowledge related to sustainability, except for the first year. For example, six practical courses (design studios) teach sustainability-related concepts over the course of a program. The study plan has only two theoretical courses, Environmental Control System (1 and 2) [48]. These courses teach students the basics of sustainability early in the second year of the program and the basic ideas behind the environmental control systems used in buildings, such as lighting, sound, and service systems. This program is a “fully integrated” model in accordance with Altomonte’s types of program structure models [44]; these models incorporate sustainability knowledge into practical and design courses and support it with related theoretical courses. According to this model, different fields come together around the main idea of the design studio project. This structure is often used in design studios as a main area that includes theoretical knowledge about how things affect the environment.

At King Saud University, knowledge of sustainable architecture is taught through architectural technology and the environment, human comfort courses, and architectural design studios. The most available knowledge related to sustainability is offered during years 3 and 5 by 31% and 51%, respectively. The other 18% of the total knowledge was delivered in years 2 and 4 (5% and 13%, respectively). Three design studios teach sustainability-related concepts during the second year and the graduation year of the program. Table 4 shows that this program utilized a combination of Altomonte models, with a “fully integrated” model implemented in year 5, a “partially integrated” model in year 3, and a linear model utilized in years 2 and 4, with five sustainability-related theoretical courses taught in a “parallel” fashion during these 2 years [47].

Other architecture schools, including Umm Al-Qura University, Effat University, and King Fahd University of Petroleum and Minerals, allocate a similar percentage (16–17%) of course contact hours to sustainability [49, 50, 57]. In contrast, Dar Al-Uloom University places marginal emphasis on sustainability, dedicating only 11% of course contact hours to this subject. The prioritization of practical courses in sustainability is minimal, with two partially sustainability-related design studio courses at Dar Al-Uloom University [52].

The study plans for the chosen schools, which can be found on their websites, show that sustainability-related elective courses were not very popular and were not doing enough to compensate for the lack of sustainability knowledge in these schools’ study programs. All three groups of students offered elective courses in varying proportions at all the programs except Dar Al-Uloom University, where no elective courses related to sustainability were available. While Effat University offers two courses, approximately 22% of its total courses relate to sustainable knowledge (EU 2023). Moreover, King Saud University, Imam Abdul Rahman bin Faisal University, and King Fahd University of Petroleum and Minerals have relatively lower values—approximately 15%, 13%, and 12%, respectively. Respectively [47–50, 52, 57].

Table 4 The integration model of curriculum in selected architecture schools in Saudi Arabia (KSA) based on Altomonte’s Program Structure Modes

Institutions	Year 1	Year 2	Year 3	Year 4	Year 5
Imam Abdulrahman Bin Faisal University (IAU)	*	<i>Partially Integrated</i>	Fully Integrated	Fully Integrated	Fully Integrated
Umm Al-Qura University (UQU)	Linear model	Linear model	<i>Partially Integrated</i>	Linear model	<i>Partially Integrated</i>
King Fahd University of Petroleum and Minerals (KFUPM)	*	*	*	Linear model	Fully Integrated
King Saud university (KSU)	*	Linear model	<i>Partially Integrated</i>	Linear model	Fully Integrated
Effat University (EU)	*	Linear model	<i>Partially Integrated</i>	Linear model	Linear model
Dar Al-Uloom University (DAU)	*	*	Linear model	<i>Partially Integrated</i>	<i>Partially Integrated</i>

*No sustainability-related knowledge

A desk survey showed that most of the chosen institutions used a “linear/parallel” model for most of the study years. The only program that does not use this model is the architecture program at Imam Abdulrahman bin Faisal University. This program embraces a “fully integrated” approach starting in the third year, continuing throughout the program, and a “partially integrated” model in the second year. However, during the graduation year, most programs follow a “fully integrated” or “partially integrated” model structure, emphasizing the application of integrated information gained during the study plan in the graduation project, including sustainability. Effat University stands out for utilizing only a “partially integrated” approach in the third year [47–50, 52, 57].

The findings reveal a variation in the integration of sustainability knowledge across the architectural education of the six selected schools in Saudi Arabia. Generally, the review indicates that most institutions have, to date, incorporated limited sustainability knowledge into their programs, primarily focusing on theoretical principles of environmental design. Additionally, there is an absence of a standardized approach to integrating sustainability knowledge into curricula, with each university adopting its own unique method. This lack of uniformity impacts the overall orientation toward sustainable design. Future research could involve surveys to gauge students’ perceptions and assess the effectiveness of sustainable architectural education, considering curriculum, skills, and performance. The goal of these kinds of studies is to create a clearer framework for teaching sustainable architecture by determining how well students understand and use sustainable principles in design studios.

5 Recommendations

Drawing from the findings of this study, the following are proposed recommendations for enhancing the integration of sustainability into architectural education in Saudi Arabia:

- *Curricular Revision:* Architectural schools should undertake a comprehensive revision of their curricula to ensure that sustainability is a core component, not just an elective or secondary subject. This should involve embedding sustainability into the DNA of all courses, from theory to technology and design studios.
- *Practical Application:* Shift the focus in design studios toward practical applications of sustainability. Encourage students to apply sustainable principles in their projects, ensuring that theoretical knowledge translates into practical skills.
- *Sustainability Metrics:* Integrate sustainability metrics and assessment tools into project evaluations to reinforce the importance of measuring environmental impact in design work.
- *Fully Integrated Model Adoption:* The fully integrated model proposed by Altomonte [44] is adopted to ensure comprehensive and seamless inclusion of sustainability principles throughout the educational experience. This model weaves sustainability into the core of all learning outcomes and project work rather than treating it as an add-on or separate component.
- *Elective/Minor Model Expansion:* Expand the Elective/Minor Model proposed by Altomonte [44] to create a more robust selection of sustainability-focused electives. These should be designed to complement the core curriculum and provide pathways for advanced study of sustainability.
- *Policy Engagement:* Engage with policymakers to align architectural education with national sustainability goals, such as those outlined in Saudi Vision 2030, thereby ensuring that educational reforms support broader environmental and societal objectives.
- *Adopting Global Standards:* Saudi Arabia should adopt the RIBA or NAAB validation procedure, which is widely recognized in global architecture education, to ensure high standards and international alignment. Registration with the Saudi Council of Engineers remains essential, emphasizing experience and qualifications for architects. This adoption will enhance professional credibility, facilitating integration into the global architectural community and fostering robust interaction between architectural education and practice.

By implementing these recommendations, architectural schools in Saudi Arabia can play a pivotal role in shaping the future where sustainable design is the norm, not the exception, and where architects are not only designers of buildings but also stewards of the environment.

6 Conclusion

In response to global challenges such as climate change and the ambitious goals of Saudi Vision 2030, the architectural sector is evolving, necessitating architects to embrace sustainable environmental design as a core competency. Architectural education plays a pivotal role in this paradigm shift, not only as an academic pursuit but also as a global imperative. This study employs a mixed-method approach, incorporating both quantitative and qualitative methodologies, with a predominant focus on quantitative analysis to examine the current state of integration of knowledge on sustainability in the curricula, the number of hours dedicated to sustainability education, the typology of courses, and the distribution of courses by academic year, focusing on national or internationally accredited institutions. The analysis reveals limited sustainability knowledge in the curricula and varying levels of emphasis on sustainability among programs, with significant disparities in the teaching hours dedicated to this critical area. In contrast, Imam Abdul Rahman Al-Faisal University has 42% of total contact hours dedicated to sustainability, while other universities have contact hours ranging from 25 to 11%. There are significant disparities in terms of emphasis, hours dedicated, and course typology. This study underscores the need for integrating sustainability across design studio levels to equip students with practical skills across disciplines. Most programs employ a "linear/parallel" or "partially integrated" model, requiring students to apply sustainability principles in their graduation projects. Despite limited sustainability knowledge in curricula, elective courses related to sustainability lack popularity and fail to address gaps. Future research should explore student perceptions and qualitative aspects to refine sustainable architectural education frameworks in Saudi Arabia, ensuring alignment with global standards and fostering comprehensive sustainability literacy among future architects.

Author contributions Mohammed Alosan has made substantial contributions to all aspects of the research and manuscript preparation. This includes conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—re-view and editing, visualization, supervision, project administration, and funding acquisition.

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Data availability Most of the data supporting the findings of this study are publicly available on the institution's website, as detailed in the references section of this manuscript. Any additional data are available upon request.

Declarations

Competing interests The authors declare no competing interests.

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Appendix A

The table below is a detailed summary of the six selected architecture programs in Saudi Arabia, specifically focusing on how these programs integrate sustainability into their curriculum.

The table within the document outlines the progression of architecture courses across five academic years, indicating the number of courses offered each year, total program credit hours, and the credit hours dedicated to core, sustainability, and sustainability-related courses. Additionally, the table includes the vision and mission statements of the institutions, which contextualize their educational objectives within the broader themes of sustainability, ethical responsibility, and alignment with Saudi Arabia's Vision 2030.

Each university's section in the table seems to provide specific data, such as:

- Total credit hours: the sum of hours for the entire architecture program.
- Programme core and sustainability course details: names of the courses categorized by their relevance to sustainability (core, partly related, etc.).
- Contact hours: the in-class time the students spend under instruction for each course.
- Practical contact hours: time spent in practical sessions, workshops, or labs.
- Sustainable credit hours (Sustain. Ch.) and contact hours (Sustain. CT): Metrics that specifically track the sustainability aspect within the courses.

The table also lists the accreditation bodies for each university, such as the National Architectural Accrediting Board (NAAB) or the National Commission for Academic Accreditation and Assessment (NCAAA), ensuring that the programs meet certain quality standards.

Comprehensive Overview of Architecture Program Coursework and Sustainability Integration at Accredited Saudi Arabian Universities

Instantiation	1 st year		2 nd year		3 rd year		4 th year		5 th year		Total program		Total Program Core Courses		Sustainability Core Courses		Sustainability Partly Related Courses		Vision and Mission	Sustainable and related courses names
	To. of courses	13	To. of courses	14	To. of courses	11	To. of courses	10	To. of courses	9	Crd. Hrs	Cont. Hrs	Credit Hours	Cont. Hrs	Cont. Hrs	%	Cont. Hrs	%		
Imam Abdulrahman Bin Faisal University IAU Accredited By NAAB, and NCAAA	To. of courses	13	To. of courses	14	To. of courses	11	To. of courses	10	To. of courses	9	171	260	134	187	59	31.5	12	6.3	Vision design and construction of sustainable built and urban environments	ARCH 201 DESIGN III ARCH 221 Env. Control (Thermal) ARCH 301 DESIGN V ARCH 324 Env. Cost. Sys. ARCH 302 DESIGN VI ARCH 401 DESIGN VII ARCH 402 DESIGN VIII ARCH 502 DESIGN X
	Contact Hours	70	Contact Hours	49	Contact Hours	48	Contact Hours	49	Contact Hours	44										
	Pr. cost hrs.*	8.2	Pr. cost hrs.*	45.12	Pr. cost hrs.*	48.11	Pr. cost hrs.*	47.9	Pr. cost hrs.*	42.8										
	Credit Hours	34	Credit Hours	34	Credit Hours	37	Credit Hours	35	Credit Hours	31										
	Sustain. Ch.*	0	Sustain. Ch.*	8.3	Sustain. Ch.*	10.4	Sustain. Ch.*	12.4	Sustain. Ch.*	12										
	Sustain. CT #	0	Sustain. CT #	4.2	Sustain. CT #	5.3	Sustain. CT #	6.6	Sustain. CT #	6										
Umm Al-Qura University UQU Accredited By ASTIN	To. of courses	10	To. of courses	14	To. of courses	14	To. of courses	13	To. of courses	20	180	232	149	199	28	14.1			None	S011201-3 Building Materials S011211-2 Environmental Control S01211-2 Urban Design S01215-4 Architectural Design (6) S01226-5 Housing S012141 Architectural Design Studio (9) S011341-5 Sustainability & Green Architecture
	Contact Hours	40	Contact Hours	44	Contact Hours	45	Contact Hours	42	Contact Hours	61										
	Pr. cost hrs.*	32.8	Pr. cost hrs.*	44.14	Pr. cost hrs.*	43.13	Pr. cost hrs.*	30.8	Pr. cost hrs.*	50.15										
	Credit Hours	30	Credit Hours	34	Credit Hours	35	Credit Hours	32	Credit Hours	49										
	Sustain. Ch.*	2	Sustain. Ch.*	2	Sustain. Ch.*	2+5	Sustain. Ch.*	2	Sustain. Ch.*	5+2										
	Sustain. CT #	2	Sustain. CT #	2	Sustain. CT #	2+10	Sustain. CT #	2	Sustain. CT #	10+2										
King Fahd University of Petroleum & Minerals KFUPM Accredited By NCAAA	To. of courses	12	To. of courses	12	To. of courses	11	To. of courses	11	To. of courses	10	171	261	101	160	13	8.1	11	6.9	Mission Increasing awareness about the social, ecological, and material realms of architecture to provide healthy, sustainable, and humane built environment within an ethically responsible framework aligned with vision 2030 of the kingdom	ARC 355 Human Factors in Architecture ARE 322 Building Mechanical Systems ARC 358 Principles of Sustainable Design ARE 328 Architectural Acoustics & Vibration ARC 402 Architectural Design Studio VII
	Contact Hours	60	Contact Hours	52	Contact Hours	54	Contact Hours	51	Contact Hours	44										
	Pr. cost hrs.*	0	Pr. cost hrs.*	30.6	Pr. cost hrs.*	49	Pr. cost hrs.*	43	Pr. cost hrs.*	35										
	Credit Hours	31	Credit Hours	36	Credit Hours	35	Credit Hours	35	Credit Hours	34										
	Sustain. Ch.*	0	Sustain. Ch.*	0	Sustain. Ch.*	0	Sustain. Ch.*	3.5	Sustain. Ch.*	10										
	Sustain. CT #	0	Sustain. CT #	0	Sustain. CT #	0	Sustain. CT #	3.5	Sustain. CT #	5										
King Saud university KSU Accredited By NAAB	To. of courses	10	To. of courses	15	To. of courses	15	To. of courses	15	To. of courses	11	171	254	121	179 (47)	35	19.6	4	2.4	Mission towards sustainability of cultural, environmental and technological goals of the department	ARCH 289 Site Analysis & Landscape ARCH 355 Studies in Environmental Control ARCH 358 Architectural Design 3 ARCH 341 Man and the Built Environment (2) ARCH 424 Facility Programming ARCH 427 Lighting and Acoustics ARCH 430 Architectural Design - 7 Arch 492 - Graduation Project
	Contact Hours	59	Contact Hours	57	Contact Hours	48	Contact Hours	50	Contact Hours	40										
	Pr. cost hrs.*	50	Pr. cost hrs.*	47.10	Pr. cost hrs.*	42.12	Pr. cost hrs.*	50.15	Pr. cost hrs.*	40.10										
	Credit Hours	32	Credit Hours	38	Credit Hours	37	Credit Hours	37	Credit Hours	34										
	Sustain. Ch.*	0	Sustain. Ch.*	2	Sustain. Ch.*	2+8.2	Sustain. Ch.*	2+3	Sustain. Ch.*	10+10										
	Sustain. CT #	0	Sustain. CT #	2	Sustain. CT #	2+4.2	Sustain. CT #	2+3	Sustain. CT #	5+5										
Effat University EU Accredited By NAAB, and NCAAA	To. of courses	12	To. of courses	14	To. of courses	14	To. of courses	12	To. of courses	9	171	247	127	184	21	11.4	6	3	Vision: With respect to environmental, social and economic sustainability.	ARCH 240 Architecture, Culture, and Environment ARCH 342 Urban Design ARCH 358 Architectural Design 3 ARCH 402 Architecture Design Studio Global Awareness ARCH 437 Energy and Economics ARCH 438 Architectural Design - 7 ARCH 492 - Graduation Project
	Contact Hours	55	Contact Hours	50	Contact Hours	46	Contact Hours	49	Contact Hours	47										
	Pr. cost hrs.*	50	Pr. cost hrs.*	54	Pr. cost hrs.*	47	Pr. cost hrs.*	46	Pr. cost hrs.*	40										
	Credit Hours	34	Credit Hours	35	Credit Hours	34	Credit Hours	34	Credit Hours	31										
	Sustain. Ch.*	0	Sustain. Ch.*	3	Sustain. Ch.*	3+10	Sustain. Ch.*	4+4	Sustain. Ch.*	3										
	Sustain. CT #	0	Sustain. CT #	3	Sustain. CT #	3+5	Sustain. CT #	3+3	Sustain. CT #	3										
Dar Al-Uloom University DAU Accredited By NCAAA	To. of courses	10	To. of courses	14	To. of courses	14	To. of courses	12	To. of courses	9	172	258	121	202	2	1	23	11.4	Mission: Engages in applying high technology, sustainable development, research excellence, community service and innovation.	ARC 404 - Environmental Control ARCH 312 - Architecture of the Arabian Region ARCH 411 - Comprehensive Design Studio II
	Contact Hours	41	Contact Hours	58	Contact Hours	54	Contact Hours	53	Contact Hours	40										
	Pr. cost hrs.*	16	Pr. cost hrs.*	53	Pr. cost hrs.*	47	Pr. cost hrs.*	46	Pr. cost hrs.*	40										
	Credit Hours	30	Credit Hours	39	Credit Hours	39	Credit Hours	37	Credit Hours	27										
	Sustain. Ch.*	0	Sustain. Ch.*	0	Sustain. Ch.*	2+3	Sustain. Ch.*	10	Sustain. Ch.*	10										
	Sustain. CT #	0	Sustain. CT #	0	Sustain. CT #	2+3	Sustain. CT #	3	Sustain. CT #	8										

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